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INFO 4220 – Networks II: Market Design
Spring 2023 – Prelim 2

Instructions:

- **Put your name and NetID in every page (do it now). Exams will be scanned and processed using gradescope. If you don't put your name and NetID in the designated spaces, we may never be able to link your answers to you.**
- **Please do not separate the pages. If you do, staple them back at the end**
- **Time limit:** You have **65 minutes** to complete this exam.
- **Asking questions:** Read all the questions now and make any questions you may have at the beginning to avoid disturbing others. If you think something is wrong or some information is missing, make the assumptions that you feel are necessary to answer and state them along with your answer.
- **Cheat Sheet:** Only this printed exam and a 1-page cheat sheet are allowed in your table during the exam. Only 1 side of the cheat sheet can be written (handwritten, not printed), your name and netid must be written on top of the cheat sheet, your cheat sheet must be your own creation, and you must turn in the cheat sheet with the exam.
- **Devices:** You cannot use any device (cell phone, smart watch, calculator, computer, etc.) during the exam.
- **Academic Integrity:** Any academic integrity violation will result in an academic integrity hearing in which a penalty consistent with Cornell's code of academic integrity will be determined.
- **Write clearly and show all your steps.** If you need more space for a question, use the last page (indicating what question the answer corresponds to)

Problem 1 (15 points, 3 points each)

1.1. In the Top Trading Cycles and Chains algorithm, why do we need chain selection rules? What do the chain selection rules need to specify?

Because TTCC is a family of algorithms. In a TTCC problem there could be multiple chains and we need a way to choose between them. A chain selection rule must specify:

- Which chain to select when there are multiple chains
- What to do with the first donor kidney: Leave it in the problem and make it available for future chains, or remove it and give it to someone in the waiting list

1.2 Explain what are the conditions that lead to efficiency in an auction (any type of auction for 1 item), and why efficiency leads to revenue maximization. Hint: Write down the social surplus of an auction and explain its components.

$$Surplus = p + (v - p) + (n - 1) * 0 = v$$

The equation above is the surplus generated by the auction. The first term (p) is what the seller receives. The second term is the surplus of the buyer (v-p). The last term (0) is the surplus received by those that don't win the auction. It is straightforward to see surplus is maximized with the bidder with

the highest valuation wins the auction. For this to happen we need that the bidder with the highest valuation bids the highest amount. For this to happen we need that bidding strategies to be symmetrical (two bidders with the same valuation bid the same amount) and strictly increasing in valuation (the highest the valuation, the highest the bid).

1.3 Why cannot the top trading cycle algorithm be readily used in the kidney exchange problem?

Because we need a way to incorporate cadaveric donors and a waiting list. Cadaveric donors are not a "stock" of organs, and thus we cannot treat them as an object in an assignment problem. They become available at random times. We need the waiting list because there may be patients that are not compatible with any of the existing donors.

1.4.a) Antonio, Bruno, and Camilo need to enroll in two courses among X, Y, and Z. Each course has 2 seats available.

The college they attend uses a bidding mechanism. Assume that the bids they submitted are as in the table below and find the courses each gets enrolled in.

	X	Y	Z
Antonio	60	38	2
Bruno	48	22	30
Camilo	47	28	34

Antonio: X, Y

Bruno: X, Z

Camilo: Z, Y

1.4.b) Suppose that now the university decided to start using the Gale-Shapley Pareto-Dominant Market Mechanism. Find the courses each student gets if they submit the same bids as in part (a), and their true preferences for courses are as follows:

Antonio	Bruno	Camilo
Z	Y	Y
X	X	X
Y	Z	Z

Antonio, Bruno, and Camilo propose to course X. Camilo has the lowest bid, thus gets rejected. After getting rejected Camilo proposes to Z

Course Y receives only 2 proposals (thus does not reject anyone): Bruno, Camilo

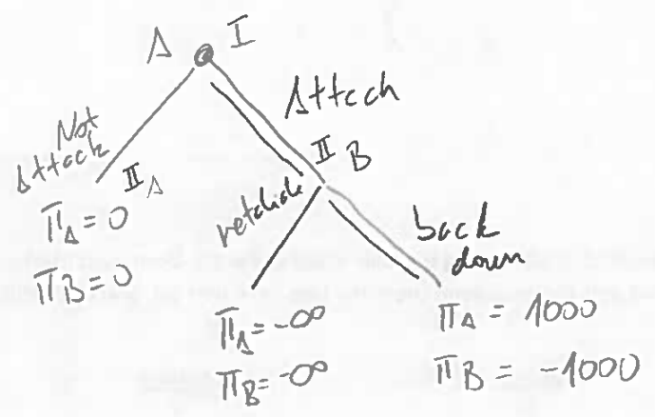
Course Z receives only 2 proposals (thus does not reject anyone): Antonio, Camilo

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Problem 2 (10 points)

There are two countries with nuclear weapons engaged in a conflict. Let's call country A the aggressor, and country B the defending country. Let's consider this is a sequential game in which A moves first and chooses between attacking or not attacking country B. If A does not attack, the game ends and both countries obtain a payoff of 0. If country A attacks, country B can retaliate or back down. If country B retaliates, a nuclear war ensues and both countries obtain a payoff if minus infinite. If country B backs down, country A obtains a payoff of 1000, and country B obtains a payoff of -1000.

a) Represent the game in extensive form (3 points)



b) Represent the game in normal form (3 points)

	Retaliate	Back Down
Not Attack	0,0	0,0
Attack	$-\infty.-\infty$	$1000,-1000$

c) Is the outcome (Not Attack, Retaliate) a Nash Equilibrium? Is it a subgame perfect equilibrium? Explain why (4 points)

(Not Attach, Retaliate) is a Nash Equilibrium because if A does not attach, B does not want to change its strategy (of retaliating), and if B is going to retaliate, A prefers not to attack.

However, it is not SGPE because retaliating is not a N.E. in the subgame that starts in node II_B. If the game ever reaches that state, the best strategy for country B would be to back down as they would get a payment of -1000 instead of $-\infty$

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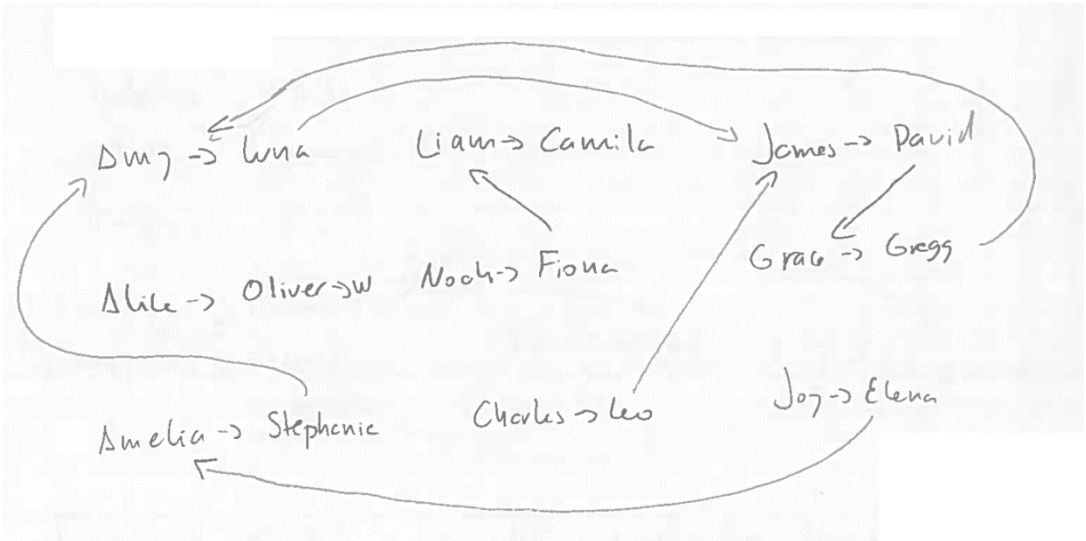
Problem 3 (13 points)

Suppose we have 10 patient-donor pairs for a kidney exchange situation: (Luna, Amy), (Camila, Liam), (David, James), (Oliver, Alice), (Fiona, Noah), (Gregg, Grace), (Stephanie, Amelia), (Leo, Charles), and (Elena, Joy). The patients have preferences regarding the donor kidneys, where "w" represents the waiting list:

Luna	**Camila**	**David**	**Oliver**	**Fiona**	**Gregg**	**Stephanie**	**Leo**	**Elena**
James	Alice	Grace	W	Liam	Amy	Amy	James	Amelia
Charles	Joy	Noah		Alice	W	Joy	Amelia	W
W	Noah	W		W		Alice	Noah	
	W					W	Liam	
							W	

a) Apply the top trading cycles and chains algorithm with the following chain selection rule: choose the longest chain and remove the patients and kidneys in that chain from the problem. Find the assignment for donors and patients. Please show your work by drawing the relevant diagrams, and show and explain each step of the algorithm. (8 points)

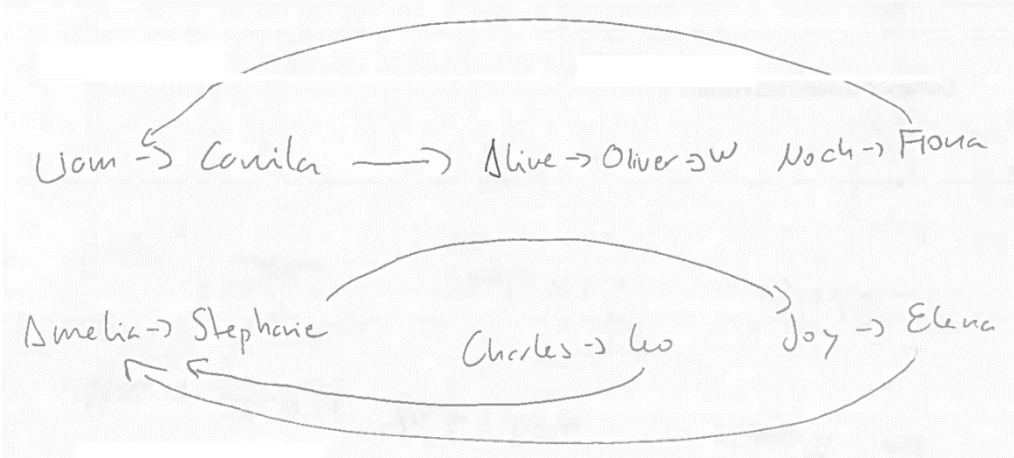
Step 1:



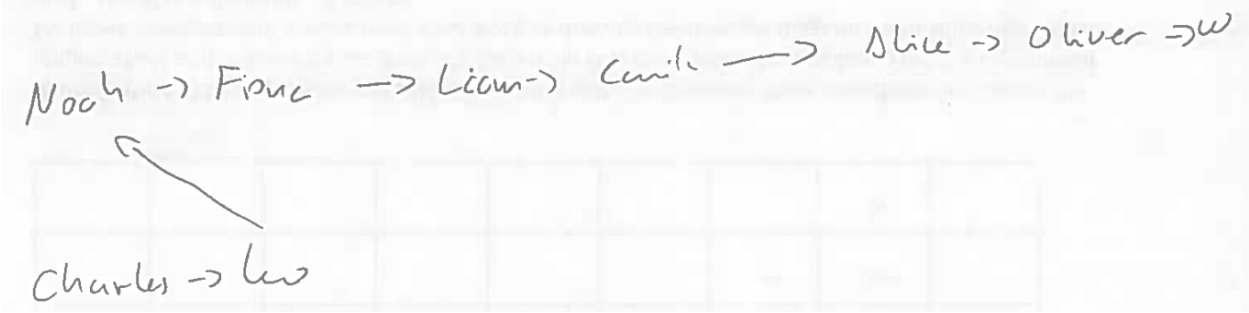
Amy – Luna – James – David – Grace – Gregg form a cycle. Lunes gets James' kidney; David gets Grace's kidney, and Gregg gets Amy's kidney. Note that when there is a cycle, we don't remove any chain that may also exist, as those patient-donors could be part of a cycle in the next step and removing them as part of a chain would be worst.

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Step 2:



Amelia – Stephanie – Joy – Elena form a cycle: Elena gets Amelia's kidney, and Stephanie gets Joy's kidney.



All the remaining patient-donors form one chain. Charles' kidney goes to someone in the waiting list, Leo gets Noah's kidney, Fiona gets Liam's kidney, Camilo gets Alice's kidney, and Oliver is added to the waiting list.

b) Is this matching Pareto efficient? If not, suggest a Pareto efficient matching and explain your rationale. (5 points)

Typically a chain selection rule that removes the shorter chain is not efficient. This happens because the chain is not made available in subsequent steps to potential add more donors/kidney. However, in this case there was only one chain and we did not run into that issue. Thus, the matching is efficient. There is no way of making anyone better off without making someone else worse off.

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Problem 4 (12 points)

Suppose that the two collectible cards from the auction in PS3, "Vickrey" and "Clark", see an increase in value due to their rarity, and the collector that purchased the cards wants to sell them using an auction. Five bidders wish to participate in the auction, and these are their valuations for each bundle of Cards:

	Vickrey	Clark	Vickrey, Clark
Bidder 1	0	10	12
Bidder 2	10	0	12
Bidder 3	0	9	14
Bidder 4	9	0	14
Bidder 5	1	1	18

a) Compute the outcome (who gets what card and at what price) if the collector uses a VCG auction (4 points)

It is straightforward to see from the table that the assignment that is going to maximize social surplus is giving the Clark to Bidder 1, and the Vickrey to Bidder 2.

To determine how much Bidder 1 must pay we compute:

Value others get when Bidder 1 participates: 10 (Bidder 2 gets the Vickrey)

Value others get when Bidder 1 does not participate: 19 (Bidder 2 gets the Vickrey and Bidder 3 gets the Clark)

Thus, Bidder 1 must pay 9.

To determine how much Bidder 2 must pay we compute:

Value others get when Bidder 2 participates: 10 (Bidder 2 gets the Vickrey)

Value others get when Bidder 2 does not participate: 19 (Bidder 2 gets the Vickrey and Bidder 4 gets the Clark)

Thus, Bidder 2 must pay 9.

b) The collector is considering that instead of using a VCG auction, they are going to run two consecutive second price auctions. They will first auction the "Vickrey", followed by the "Clark". Assume that in the first auction, all bidders will participate considering only their valuations for the Vickrey (i.e. they will not think of complementarities or what could happen in the second auction at this time). Then, in the second auction, they will decide how much to bid based on what they got in the first auction and their valuations for bundles of cards. Compute the outcome using this method and assumptions. For each auction, explain how much each bidder bids and why (4 points)

In the first auction everyone bids their true valuation for the Vickrey. Bidder 2 gets the Vickrey and pays 9.

In the second auction, everyone bids their valuation for the Clark, except for Bidder 2 that bids 2, which is the additional value they would get from getting the Clark. Bidder 1 gets the Clark and pays 9.

c) Which method is better for the collector? Which method maximizes social value? Which method is preferred by the bidders (thinking of all bidders collectively)? (4 points)

Both methods produce the same outcome, so the collector and the bidders are indifferent. The total social value in both methods is 20 (18 for the auctioneer, and 2 for the bidders).